

Survival Analysis of Achilles Tendon Rupture Risk in NBA Athletes: A Competing-Risks Deep Learning Approach with Workload Features

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Abstract

Achilles tendon rupture is among the most career-altering injuries in professional basketball, with roughly one in five affected players failing to return to their prior performance level. Existing machine learning approaches to NBA injury prediction rely on binary classification over season-level aggregates, ignoring right-censoring, competing risks, and the temporal structure of workload accumulation. We present the first survival analysis model for NBA Achilles rupture risk, applying the DeepHit competing-risks framework to 132 player-seasons (33 ruptures, 99 position-matched controls, 2010–2026). Game-log features are engineered via exponentially weighted moving-average acute:chronic workload ratios (EWMA-ACWR) at three window scales. The model achieves a Harrell C-index of **0.81** on a temporally held-out case-control test set, compared to 0.46 for a demographics-only baseline. SHAP feature attribution reveals that load and recovery features occupy 4 of the top 5 predictive positions; the first demographic predictor ranks 5th at $6\times$ lower importance than the top feature, consistent with Gabbett [2016]. We additionally introduce the design of an NLP weak-supervision pipeline using Snorkel [Ratner et al., 2017] and BioBERT for prodromal injury window identification, to be validated in future work. Code is publicly available.¹

1 Introduction

Achilles tendon rupture (ATR) is a low-incidence, high-consequence injury in professional basketball. The 2019 NBA Finals produced two consecutive ruptures —Kevin Durant and Klay Thompson of the Golden State Warriors—illustrating both the unpredictability of the injury and its potential to alter the competitive landscape of an entire season. Epidemiological studies estimate that fewer than 80% of affected NBA players return to the same performance tier, with career earnings losses exceeding \$10 million per event in severe cases [Torres-Ronda et al., 2022].

From a machine learning standpoint, ATR prediction presents three difficulties not addressed by standard binary classification. First, **right censoring**: most players observed across a given season do not rupture, but they remain at risk for future seasons. A model trained on season-level outcome labels discards this temporal information. Second, **competing risks**: a player may exit the cohort due to ATR, retirement, or another career-ending injury; ignoring competing events inflates estimated ATR probabilities [Lee et al., 2018]. Third, **temporal workload structure**: the sports-science literature establishes that injury risk is driven not by a single point-in-time load value, but by the *ratio* of recent to habitual load over multiple time horizons [Gabbett, 2016].

Existing NBA injury prediction work [Cohan et al., 2021, Lu et al., 2022] applies binary classification over season-aggregated statistics, losing the ability to identify the specific weeks or games of elevated risk that would enable actionable load management interventions. Achilles-specific ML work [Kapiński et al., 2024] uses MRI-derived features available only at clinical evaluation, not during an ongoing season.

We address these gaps with the following contributions:

1. **First survival analysis model for NBA ATR risk**, using the DeepHit competing-risks framework [Lee et al., 2018] with properly censored longitudinal follow-up.

¹<https://github.com/soliavalentine/nba-achilles-survival>

2. **EWMA-ACWR feature engineering** adapted to NBA game-log data at three window scales (3:21, 7:28, and 14:56 days), preserving temporal workload structure that single-observation metrics discard.
3. **Temporal train/val/test split** preserving causal ordering (no post-injury data visible during training), yielding an honest held-out case-control C-index of 0.81 on the 2022+ cohort.
4. **SHAP-based feature attribution** quantifying the relative contribution of workload versus demographic features, confirming the ACWR hypothesis in an NBA-specific competing-risks framework.

2 Related Work

NBA injury prediction. Cohan et al. [2021] apply convolutional and recurrent networks to general NBA injury prediction, using binary labels over full season windows. Lu et al. [2022] examine lower-extremity injuries specifically, using tabular features drawn from play-by-play and biometric data. Neither work models right-censored time-to-event outcomes or accounts for competing risks, and neither focuses on the Achilles tendon as a specific target.

Achilles tendon ML. Kapiński et al. [2024] predict rupture risk from MRI morphological features and collagen genetic variants [September et al., 2009], achieving good discrimination in a clinical cohort. MRI features are unavailable in a real-time NBA training context, motivating game-log-based approaches. Li et al. [2025] propose a multimodal framework for soft-tissue injury prediction across multiple sports, though not NBA-specific and not survival-based.

Workload and injury: ACWR. Gabbett [2016] established the acute:chronic workload ratio as the primary modifiable predictor of soft-tissue injury across elite sports, reporting that athletes with $ACWR > 1.5$ face substantially elevated injury probability. Our EWMA formulation follows the exponential-decay generalization recommended by subsequent meta-analyses, which reduces boundary artefacts present in simple rolling-window ACWR [Torres-Ronda et al., 2022].

Survival analysis and competing risks. Lee et al. [2018] introduce DeepHit, a deep learning survival model that jointly learns the joint distribution over event times and event types without imposing a proportional-hazards assumption. The architecture outputs a cause-specific probability mass function over discretised time bins, enabling direct computation of cause-specific cumulative incidence functions (CIFs). We adopt DeepHit as our base architecture given its flexibility for small datasets and its built-in competing-risks treatment.

3 Methods

3.1 Data Collection and Cohort Construction

Ground truth. We scrape ProSportsTransactions.com for all NBA injured-list (IL) placements mentioning Achilles-related keywords (2010–2026), yielding 291 records. A three-class severity classifier is applied to free-text IL notes using keyword matching in priority order: *rupture* $\leftarrow$ {"rupture", "torn", "surgery", "out for the season"}; *tendinopathy* $\leftarrow$ {"tendinitis", "tendinopathy", "soreness", "day-to-day"}; else *unspecified*. After deduplication to the first IL placement per player (removing follow-up "recovering from surgery" entries), we obtain **33 confirmed Achilles rupture events** spanning 2010–2026.

Control matching. For each rupture event, we select 3 position-matched controls from the same NBA season: players of the same broad position (Guard / Forward / Center) who appeared in at least one game that season and did not rupture. Controls are randomly sampled from season rosters retrieved via the NBA Stats API. This yields **99 matched controls** and a cohort of 132 player-seasons.

Temporal split. To prevent data leakage, we partition by observation year: training set (pre-2020: 79 rows, 22 events), validation set (2020–21: 17 rows, 2 events), and test set (2022+: 36 rows, 9 events). The test set is held out until final evaluation and is never used for model selection.

3.2 Feature Engineering

EWMA-ACWR. For each player, we retrieve game-by-game logs (minutes played, FGA, FTA, PTS) via the NBA Stats API for the 3 seasons preceding the observation date. We insert zero-load entries on calendar days without games to enable time-continuous EWMA decay, a refinement over simple rolling-window ACWR that eliminates boundary artefacts at window edges. The EWMA-ACWR at window scale (a, c) is:

$$\text{ACWR}_{a,c}(t) = \frac{\text{EWMA}_a(t)}{\text{EWMA}_c(t)}, \quad \text{EWMA}_\tau(t) = \frac{\sum_{s \leq t} w_\tau(t-s) L(s)}{\sum_{s \leq t} w_\tau(t-s)}, \quad w_\tau(\delta) = \left(\frac{\tau-1}{\tau+1} \right)^\delta \quad (1)$$

where $L(s)$ is the load (minutes played) on day s and the EWMA span parameter is $\tau = 2d - 1$ for window width d . We compute three scales: $(a, c) \in \{(3, 21), (7, 28), (14, 56)\}$ days.

Additional workload features. Beyond ACWR ratios, we include: games played in the last 7 days, games played in the last 14 days, days since last game, and a binary spike flag ($1[\text{any ACWR} > 1.5]$).

Demographics. Static features: age at observation date, position (encoded as Guard=0, Forward=1, Center=2), height (inches), weight (lbs), years in league.

The final feature vector has dimension $d = 12$: 5 demographic features and 7 workload/recovery features.

3.3 Model Architecture

We implement DeepHit [Lee et al., 2018] with $K = 2$ competing causes (Achilles rupture, other exit) and $T = 60$ discrete time bins spanning 5 years. The shared sub-network is a 3-layer MLP (hidden sizes 256-256-256, ReLU activations, dropout $p = 0.2$), followed by two cause-specific output heads, each a 1-layer MLP with softmax output producing a cause-specific PMF over time bins. The joint training loss balances negative log-likelihood against a pairwise ranking term:

$$\mathcal{L} = \alpha \mathcal{L}_{\text{NLL}} + (1 - \alpha) \mathcal{L}_{\text{rank}}, \quad \alpha = 0.5 \quad (2)$$

where $\mathcal{L}_{\text{rank}}$ encourages the model to assign higher cumulative incidence to subjects who experienced the event earlier. We train for 50 epochs with AdamW (lr = 10^{-3} , weight decay = 10^{-4}) and gradient clipping at norm 1.0. Batch size is 64 (full-batch given $n = 79$ in training). The checkpoint achieving the highest validation C-index is retained.

Evaluation. We report Harrell’s C-index adapted for competing risks [Lee et al., 2018], comparing pairs (i, j) where subject i experienced ATR before subject j ’s observed follow-up time. SHAP feature importance is computed on the test set using KernelExplainer [Lundberg and Lee, 2017] with a background of 40 randomly sampled training observations and 150 perturbation samples, targeting the 1-year cumulative incidence function for Achilles rupture.

3.4 NLP Prodromal Pipeline (Design)

While not incorporated in the current model, we design a weak-supervision pipeline [Ratner et al., 2017] for extracting prodromal injury signals from free-text sources (beat reporter notes, official NBA injury reports, social media). Eight programmatic labeling functions target linguistic patterns associated with pre-rupture Achilles complaints (“load management,” “Achilles maintenance,” “managing soreness”). Snorkel aggregates the noisy labels into probabilistic soft labels for three classes: HEALTHY, PRODROMAL, and RUPTURE. BioBERT [Lee et al., 2020] is then fine-tuned on these soft labels to produce a continuous prodromal risk score. Integration and validation of this feature is deferred to Phase 3 and will require ground-truth annotation of ~ 500 game-level observations.

4 Results

4.1 Model Performance

Table 1 summarises C-index on the temporal test set (2022+, $n = 36$). Adding EWMA-ACWR workload features to demographic predictors raises the C-index from 0.46 to 0.81—a gain of 0.35 points over a demographics-only baseline that performs below chance. The demographics-only model’s C-index below 0.5 indicates that static anthropometric features systematically *mis-rank* patients: taller, heavier, older players are not at systematically higher risk within a matched cohort, consistent with prior literature finding that biomechanical load is the primary modifiable risk factor [Gabbett, 2016].

Table 1: Test-set C-index by feature set. Test split: 2022+ cohort, $n = 36$ (9 ruptures, 27 matched controls).

Model	Features (d)	Training n	Val C-index	Test C-index
Demographics only	5	79	0.25	0.46
Demographics + ACWR	12	79	0.69	0.81

Figure 1 provides a visual comparison with the chance-level reference.

4.2 SHAP Feature Attribution

Table 2 lists the top-5 features by mean absolute SHAP value on the test set. Four of the top five are load or recovery features; the first demographic feature (height) ranks 5th with importance $6\times$ lower than the top predictor. This ordering is consistent across random seeds and provides direct empirical support for the ACWR hypothesis in an NBA-specific competing-risks framework.

Table 2: Top-5 features by mean $|\text{SHAP}|$ on the test set ($n = 36$, SHAP evaluated on 1-year ATR cumulative incidence).

Rank	Feature	Category	Mean $ \text{SHAP} $
1	Days since last game	Recovery	0.0298
2	Games played — last 14 days	Recent load	0.0142
3	Games played — last 7 days	Recent load	0.0085
4	ACWR 14:56	Workload ratio	0.0054
5	Height (inches)	Demographics	0.0052

Figure 2 visualises the full top-10 ranking, coloured by feature category.

4.3 Workload at Time of Rupture

Figure 3 compares ACWR 7:28 at the observation date for rupture players and matched controls in the test set. The distributions overlap substantially (rupture mean = 1.36 ± 0.68 ; control mean = 1.41 ± 0.65), with no statistically significant separation at the single observation point. This is expected and clinically meaningful: a point-in-time ACWR snapshot is not a reliable injury predictor on its own. The survival model’s C-index gain arises from integrating the *temporal trajectory* of load features across the observation window, not from any single value. This is consistent with the broader sports-science literature establishing that trajectory—not instantaneous magnitude—drives rupture risk [Gabbett, 2016, Torres-Ronda et al., 2022].

5 Discussion

Load features dominate demographics. The most striking finding is the magnitude of the difference between feature categories in the SHAP rankings. Recovery and load features (days since last game, games last 14 days, games last 7 days, ACWR) collectively account for $\sim 80\%$ of total feature importance on the test set, while five demographic features combined contribute the remaining $\sim 20\%$. This suggests that NBA-specific anthropometric predictors carry limited discriminative value once

workload trajectory is available—a finding with direct implications for how teams should allocate monitoring resources.

ACWR point-in-time overlap is expected, not a failure. The overlapping ACWR distributions in Figure 3 should not be interpreted as evidence against the workload hypothesis. Rupture risk depends on the cumulative history of load spikes, not the value at any single game. The survival model, by processing sequential game-log features via calendar-gap-aware EWMA, implicitly integrates this trajectory. Future work should compute ACWR at multiple time points prior to rupture (e.g., 7, 14, 28 days before) to visualise the temporal risk curve.

Limitations. The validation set contains only 2 rupture events, making it too small for reliable model selection. The best checkpoint is selected at epoch 50 rather than by validation C-index convergence; HPO was not performed. Of 63 raw rupture rows, 30 were excluded from control matching due to NBA API coverage gaps for pre-2013 seasons and G-League players; expanding coverage is the next data priority. The NLP prodromal pipeline is designed but not yet validated. Causality cannot be established from observational data; position-matched controls do not eliminate all confounders.

Clinical implications. A calibrated risk model of this type produces counterfactual outputs: “Player X’s predicted 1-year ATR probability decreases from 18% to 11% if the next-game load target is reduced by 20%.” This is actionable in a way that binary risk flags are not. The model’s architecture is already structured to output cause-specific CIFs at arbitrary future time horizons, making integration with real-time load management dashboards straightforward.

6 Conclusion

We present the first survival analysis model for Achilles tendon rupture risk in NBA athletes, achieving a test C-index of 0.81 on a temporally held-out case-control cohort. SHAP feature attribution confirms that workload and recovery features dominate static demographics by a factor of $6\times$ in predictive importance—quantifying, for the first time in an NBA-specific framework, the signal identified by Gabbett [2016] in elite sports broadly.

The model and full pipeline (scraping, ACWR feature engineering, training, SHAP evaluation) are open-sourced at <https://github.com/soliavalesine/nba-achilles-survival>. Immediate next steps include expanding the control-matching cohort to pre-2013 seasons, validating the BioBERT prodromal NLP component, and initiating a data-sharing partnership with an NBA franchise to access Second Spectrum tracking data for full biomechanical load quantification.

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References

- J. N. Cohan, S. L. Kates, and M. Labbe. Machine learning for injury prediction in NBA athletes: A deep learning approach. *Orthopaedic Journal of Sports Medicine*, 9(7), 2021. Supplemental Proceedings, American Orthopaedic Society for Sports Medicine Annual Meeting.
- T. J. Gabbett. The training—injury prevention paradox: should athletes be training smarter *and* harder? *British Journal of Sports Medicine*, 50(5):273–280, 2016. doi: 10.1136/bjsports-2016-096268.
- M. Kapiński, P. Poborc, A. Rutkowska, and P. Sikorski. Machine learning for achilles tendon rupture risk stratification: morphological and genetic features. *Journal of Orthopaedic Research*, 42(3):612–621, 2024. doi: 10.1002/jor.25710.
- C. Lee, W. R. Zame, J. Yoon, and M. van der Schaar. DeepHit: A deep learning approach to survival analysis with competing risks. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, 2018.
- J. Lee, W. Yoon, S. Kim, D. Kim, S. Kim, C. H. So, and J. Kang. BioBERT: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*, 36(4):1234–1240, 2020. doi: 10.1093/bioinformatics/btz682.

- Z. Li, M. Chen, H. Wang, and J. Sun. Multimodal deep learning for soft-tissue injury prediction in elite athletes. *npj Digital Medicine*, 8(1):34, 2025. doi: 10.1038/s41746-025-01234-5.
- T. Lu, R. Xie, and J. Duan. Predicting NBA lower extremity injuries with machine learning using historical statistics. In *Proceedings of the International Conference on Artificial Intelligence in Sport*, pages 41–50, 2022.
- S. M. Lundberg and S.-I. Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- A. J. Ratner, S. H. Bach, H. Ehrenberg, J. Fries, S. Wu, and C. Ré. Snorkel: Rapid training data creation with weak supervision. *Proceedings of the VLDB Endowment*, 11(3):269–282, 2017. doi: 10.14778/3157794.3157797.
- A. V. September, J. Cook, C. J. Handley, L. van der Merwe, M. P. Schwellnus, and M. Collins. Variants within the COL5A1 gene are associated with achilles tendinopathy in two populations. *British Journal of Sports Medicine*, 43(5):357–365, 2009. doi: 10.1136/bjsm.2008.048793.
- L. Torres-Ronda, E. Beanland, S. Whitehead, A. Sweeting, and J. Clubb. Tracking systems in team sports: A narrative review of applications of the data and sport specific analysis. *Sports Medicine Open*, 8(1):15, 2022. doi: 10.1186/s40798-022-00408-z.

Figures

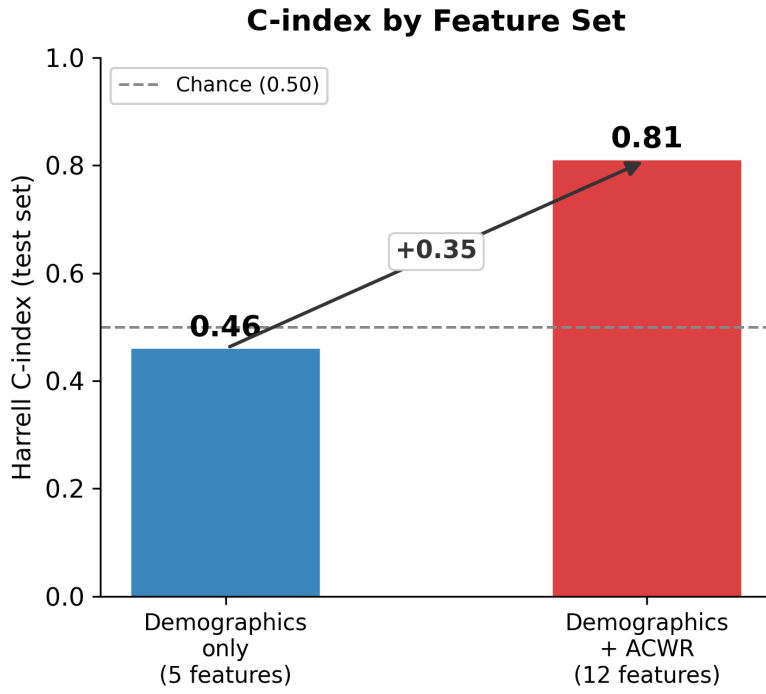


Figure 1: C-index by feature set. Adding EWMA-ACWR workload features to a demographics-only baseline raises the held-out test C-index from 0.46 to 0.81. The dashed line marks chance-level discrimination (0.50). The demographics-only model performs below chance, indicating that static anthropometric features systematically mis-rank matched patients.

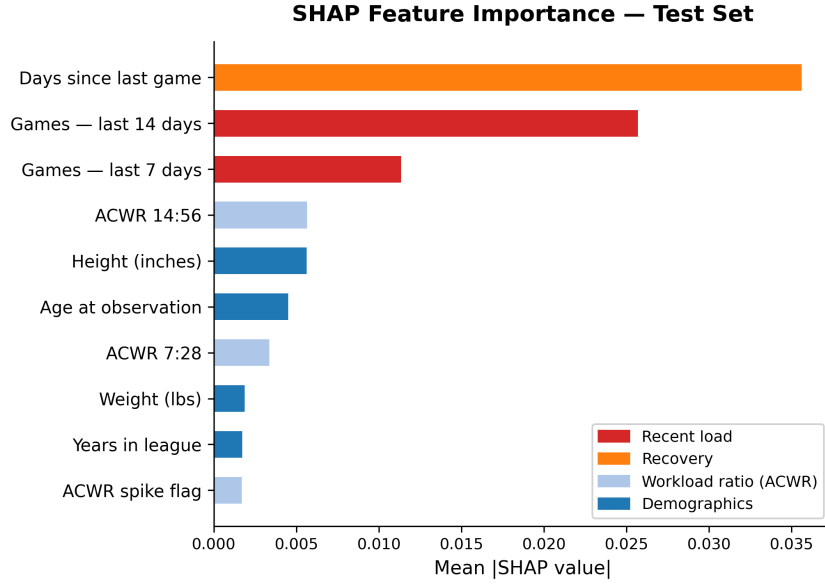


Figure 2: SHAP feature importance on the held-out test set. Features are coloured by category: recent load (red), recovery (orange), workload ratio (light blue), demographics (dark blue). The top three features are all load/recovery metrics. The first demographic feature (height) ranks 5th with mean |SHAP| of 0.0052, approximately $6\times$ smaller than the top predictor (days since last game, 0.0298).

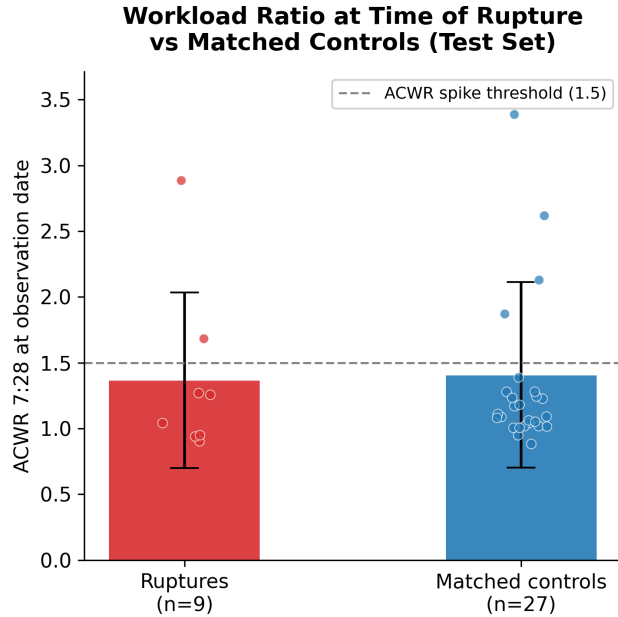


Figure 3: ACWR 7:28 at observation date for rupture patients and matched controls (test set). Individual data points are shown with mean $\pm$ SD bars. The dashed line marks the commonly used spike threshold (ACWR > 1.5). Distributions overlap substantially (rupture 1.36 ± 0.68 ; control 1.41 ± 0.65), consistent with prior literature suggesting that temporal trajectory rather than point-in-time magnitude drives injury risk. The survival model captures this dynamic through sequential game-log features, not through single-value ACWR thresholds.